The Calabi–Yau-to-chiral functor and the $\kappa_\bullet$-spectrum

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Abstract

Where do chiral algebras come from? Volumes I and II of the modular Koszul duality programme accept a chiral algebra as given and extract its invariants: bar-cobar adjunction, the modular characteristic κ_{ch}, the shadow tower, the five theorems A–D+H. The geometric source is left unspecified. This paper constructs it.

A Calabi–Yau category C of dimension d carries a negative-cyclic Calabi–Yau class $[\sigma_C] \in HC_d^-(C)$ whose Hochschild shadow is a non-degenerate cyclic pairing. On the verified specialisation locus, the CY-to-chiral assignment

$$\Phi_d^{(\Sigma_{d-1}, C)} \;=\; \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1}, C} \circ \Phi_d^{\mathrm{FA}} \colon \mathrm{CY}_d^{\mathrm{wit}}\text{-Cat} \longrightarrow E_n\text{-ChirAlg}$$

factors in two stages. Stage 1, Φ_d^{FA}, produces a holomorphic E_d-factorisation algebra $\Phi_d^{\mathrm{FA}}(C)$ on a CY_d variety X after the needed formality, locality, and completion witnesses are fixed. Its algebraic input is the Gerstenhaber bracket of degree -1 on $\mathrm{HH}^\bullet(C)$, or degree $1-d$ after CY duality to Hochschild homology. Stage 2, $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1}, C}$, is factorisation homology over a closed $(d-1)$-cycle $\Sigma_{d-1} \subset X$ followed by restriction to a reference curve $C \subset X$. The four structural operations, Lie conformal algebra $\mathfrak{L}_C$ from the Gerstenhaber bracket, factorisation envelope $\mathrm{Fact}_C(\mathfrak{L}_C)$, witnessed framing data, and negative-cyclic quantisation, realise the composite. The assignment is proved at $d = 2$ (Theorem CY-A$_2$, E_2-chiral on any curve C). At $d = 3$ it is an object-level E_1-chiral construction on the witnessed filtered locus. A single CY_d category admits a family of E_1-chiral shadows indexed by the choice of (Σ_{d-1}, C); the Borcherds–Δ_5, Igusa–χ_{10}, and paramodular Gritsenko–Cléry outputs at $d = 3$ are distinct specialisations of one canonical Φ_3^{FA}.

A single CY manifold admits multiple chiral algebraizations, each with its own modular characteristic. The resulting $\kappa_\bullet$-spectrum $\mathrm{Spec}_{\kappa_\bullet}(X) = \{\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}\}$: four numbers measuring four faces of the same manifold. For $K3 \times E$: $\mathrm{Spec}_{\kappa_\bullet} = \{0, 3, 5, 24\}$. The value 2 belongs to the K3 fibre lane, not to the total-space categorical characteristic. The shadow–Siegel gap theorem identifies four structural obstructions separating $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$ from $\kappa_{\mathrm{BKM}} = 5$.

For toric CY_3, the functor interfaces with the critical cohomological Hall algebra: $\mathrm{CoHA}(Q_X, W_X) \;\simeq\; Y^+(\widehat{\mathfrak{gl}}_{Q_X})$ provides the E_1-sector, and the Drinfeld center recovers E_2-braiding. The E_1 stabilization theorem proves that for toric $GL(d)$-invariant local models at $d \geq 3$, the relevant invariant Schouten–Nijenhuis sector is silent. General CY categories acquire their E_1 output from Stage 2 and the CY degree shift.

Contents

1 Introduction

1.1 The source problem

A chiral algebra is a solution. What is the problem it solves?

The bar-cobar machine of Volume I extracts from any chiral algebra A on a curve X a sequence of invariants: the bar complex $B(A) = T^c(s^{-1}\bar{A})$ with its deconcatenation coproduct, the universal Maurer–Cartan element Θ_A satisfying $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$, the modular characteristic $\kappa_{\mathrm{ch}}(A)$ governing the genus-1 curvature, and the shadow tower $\{S_r\}_{r\geq 2}$ encoding higher-genus data. The five theorems (A–D+H) are structural properties of this machine. Volume II supplies the physical comparison surface: the derived chiral center $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = C^\bullet_{\mathrm{ch}}(A, A)$ is the universal closed-sector cochain object of A. It becomes the bulk observable algebra of a 3d holomorphic-topological gauge theory only after the open–closed comparison map has been constructed; the $\mathrm{SC}^{\mathrm{ch,top}}$ pair $(C^\bullet_{\mathrm{ch}}(A, A), A)$ is the closed-sector–boundary algebraic system. It is a physical boundary–bulk system only after that comparison.

Neither volume constructs a single chiral algebra. Affine Kac–Moody algebras arise from flat connections; Virasoro from reparametrizations; every other family in the standard landscape must be supplied by hand. The question is: what functor connects geometric input to chiral algebraic output, and what structure must it preserve?

The answer is the CY-to-chiral specialization $\Phi_d^{(\Sigma_{d-1},C)} = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}$. A Calabi–Yau category C carries a cyclic A_∞-structure encoding a d-dimensional Frobenius condition at the chain level. The first stage Φ_d^{FA} converts this datum into a canonical E_d-holomorphic factorisation algebra on the CY_d variety X; the second stage $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}$ specialises along a closed $(d-1)$-cycle $\Sigma_{d-1} \subset X$ and restricts to a reference curve $C \subset X$, producing the chiral algebra A_C on C with three preservation properties: the bar complex of A_C recovers the cyclic bar complex of C, the modular characteristic $\kappa_{\mathrm{ch}}(A_C)$ equals the CY Euler characteristic $\chi^{\mathrm{CY}}(C)$, and the $\mathbb{S}^d$-framing determines the operadic level on C.

1.2 The $\kappa_\bullet$-spectrum

A single CY manifold admits multiple chiral algebraizations. The lattice vertex algebra V_Λ attached to $H^*(K3,\mathbb{Z})$, the generalized Kac–Moody algebra whose denominator is the Igusa cusp form Δ_5, and the Conway module from the Leech lattice frame are three algebraizations of the same K3 surface. Each carries its own modular characteristic.

The $\kappa_\bullet$-spectrum records all four lanes:

$$\begin{aligned}
\kappa_{\mathrm{cat}} &= \chi(\mathcal{O}_S) && \text{(holomorphic Euler characteristic),}\\
\kappa_{\mathrm{ch}}^{\mathrm{Heis}} &= \dim_{\mathbb{C}} X && \text{(Heisenberg–Mukai chiral specialisation),}\\
\kappa_{\mathrm{BKM}} &= \tfrac{1}{2}\,\mathrm{wt}(\Phi_{10}^{\mathrm{un}}) && \text{(Borcherds–Kac–Moody automorphic weight),}\\
\kappa_{\mathrm{fiber}} &= \mathrm{rank}(\Lambda) && \text{(lattice rank).}
\end{aligned}$$

For $K3 \times E$: $\mathrm{Spec}_{\kappa_\bullet} = \{0, 3, 5, 24\}$. The four numbers measure four faces of the same manifold: holomorphic Euler characteristic, boundary chiral algebra, Borcherds denominator weight, and lattice frame. The chiral-holographic pair $(\kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}) = (3,5)$ governs the boundary–Borcherds comparison axis of Volumes I–II. The $\kappa_\bullet$-spectrum is invisible from any single chiral algebra; it is intrinsic to the CY category.

1.3 The E_1 stabilization theorem

For $d \geq 3$, the CY chiral algebra is natively E_1, not E_d. The full Schouten–Nijenhuis bracket on $\mathrm{PV}^*(\mathbb{C}^d)$ does not vanish. The toric $GL(d)$-invariant CY sector used below is the abelian local sector, and the CoHA extension correspondence (sub before quotient) imposes a preferred ordering, which is E_1. The braided monoidal structure is recovered through the Drinfeld center of the E_1-monoidal representation category.

The obstruction to additional structure beyond E_1 is controlled by $\pi_d(BU)$, which is $\mathbb{Z}$ for even d and 0 for odd d by Bott periodicity. The effective obstruction is trivial precisely at $d \bmod 8 \in \{1,3,7\}$.

1.4 Outline

Section 2 reviews CY categories and cyclic A_∞-structures. Section 3 constructs the four-step functor Φ and proves Theorem CY-A_2. Section 4 defines the $\kappa_\bullet$-spectrum and develops its properties. Section 5 works through the $K3 \times E$ prototype in detail: the thirteen structural results K3-1 through K3-13, the shadow–Siegel gap theorem, and the four obstructions separating κ_{ch} from κ_{BKM}. Sections 6–7 treat toric CY_3, the CoHA, and the E_1 stabilization theorem.

2 Calabi–Yau categories and cyclic A_∞-structures

2.1 CY categories

Definition 2.1 (Calabi–Yau category). A smooth proper dg category C over a field k is *Calabi–Yau of dimension d* if there exists a quasi-isomorphism of C-bimodules

$$C^\vee \xrightarrow{\sim} C[d]$$

where $C^\vee = \mathrm{RHom}_{C^e}(C, C^e)$ is the inverse dualizing bimodule and $[d]$ denotes the cohomological shift by d.

The CY condition encodes Serre duality at the categorical level. For the bounded derived category $D^b(\mathrm{Coh}(X))$ of a smooth projective variety X with trivial canonical bundle $\omega_X \simeq \mathcal{O}_X$, Serre duality gives $\mathbb{S}_X \simeq [d]$ on $D^b(\mathrm{Coh}(X))$ (where $d = \dim_{\mathbb{C}} X$), and the resulting CY_d-structure is the classical Serre functor.

Example 2.2 (Standard families). Four families of CY categories provide the standard landscape:

(i) *Elliptic curves* ($d = 1$): $D^b(\mathrm{Coh}(E_\tau))$ is CY_1; the functor Φ produces the Heisenberg vertex algebra H_k at level $k = \mathrm{vol}(E_\tau)$.

(ii) *K3 surfaces* ($d = 2$): $D^b(\mathrm{Coh}(S))$ is CY_2; Φ produces an E_2-chiral algebra with $\kappa_{\mathrm{ch}} = \chi(\mathcal{O}_S) = 2$.

(iii) *CY threefolds* ($d = 3$): the conifold, local $\mathbb{P}^2$, $K3 \times E$. The functor Φ at $d = 3$ is proved (Theorem CY-A_3).

(iv) *CY fourfolds* ($d = 4$): $K3 \times K3$, the sextic fourfold. The obstruction $\pi_4(BU) = \mathbb{Z}$ controls the shifted symplectic structure.

2.2 Cyclic A_∞-structure

A CY category C of dimension d carries a negative-cyclic Calabi–Yau class whose Hochschild shadow is a non-degenerate trace:

$$\mathrm{Tr}\colon \mathrm{HH}_d(C) \longrightarrow k.$$

The negative-cyclic class is the CY datum; the Hochschild trace is its decategorified shadow. The primary chain object is the cyclic bar complex $\mathrm{CC}_\bullet(C)$ equipped with the Connes B-operator. Higher framing data are additional witnesses, not a formal consequence of the trace alone.

Definition 2.3 (Cyclic A_∞-structure). A *cyclic A_∞-structure* on a CY_d category C consists of:

(i) An A_∞-category structure on C: operations $\mu_k\colon C^{\otimes k} \to C[2-k]$ for $k \geq 1$ satisfying the Stasheff identities.

(ii) A non-degenerate cyclic pairing $\langle -, - \rangle\colon C \otimes C \to k[-d]$ satisfying the higher cyclicity conditions $\langle \mu_k(a_1, \ldots, a_k), a_0 \rangle = \pm \langle \mu_k(a_0, a_1, \ldots, a_{k-1}), a_k \rangle$.

(iii) The induced Connes B-operator $B\colon \mathrm{CC}_n(C) \to \mathrm{CC}_{n+1}(C)$ with $B^2 = 0$ and $Bb + bB = 0$.

Proposition 2.4 (Categorical Hochschild homology and the Gerstenhaber bracket)**.** *The categorical Hochschild cohomology* $\mathrm{HH}^\bullet(C)$ *of any dg category carries a Gerstenhaber bracket* $[-,-]$ *of degree* -1*. For a* CY_d *category, the CY pairing identifies* $\mathrm{HH}^\bullet(C) \simeq \mathrm{HH}_{d-\bullet}(C)$*, and the Gerstenhaber bracket becomes a Lie bracket on* $\mathrm{HH}_\bullet(C)$ *of degree* $1-d$*.*

Proof. The Gerstenhaber bracket is a standard structure on categorical Hochschild cohomology (Gerstenhaber 1963, Keller 2006); the CY identification with homology uses the non-degenerate trace and the Hochschild–Kostant–Rosenberg degeneration at the E_1-page of the Hodge-to-de Rham spectral sequence. □

2.3 The framing witness

The CY_d-structure on C supplies the negative-cyclic and Connes data. A chain-level $\mathbb{S}^d$-framing is an additional witness. The $\mathbb{S}^1$-action on $\mathrm{HH}_\bullet(C)$ is the Connes shadow; the full framing, when constructed, governs the operadic level of the chiral algebra.

Proposition 2.5 (Framing witness; Kontsevich–Vlassopoulos)**.** *For a* CY_d *category* C*, the negative-cyclic CY class supplies the Connes action on* $\mathrm{CC}_\bullet(C)$*. At* $d = 2$*, the Kontsevich–Vlassopoulos framing witness provides the* $\mathbb{S}^2$*-action needed for the* E_2*-algebra structure on cyclic homology. At* $d = 3$*, the corresponding chain-level* $\mathbb{S}^3$*-framing is part of the witnessed filtered locus.*

The framing witness is the datum that the assignment Φ consumes to determine the operadic enhancement. At $d = 2$, the $\mathbb{S}^2$-framing is proved. At $d = 3$, the chain-level $\mathbb{S}^3$-framing compatible with the BV structure is one of the explicit witnesses imposed on the constructed locus.

Remark 2.6 (Bott periodicity and the framing obstruction). The obstruction to extending the $\mathbb{S}^d$-framing beyond the base E_1 level is controlled by $\pi_d(BU)$:

d	Native E_n	$\pi_d(BU)$	Obstruction	Example
1	E_∞	0	trivial	elliptic curve
2	E_2	$\mathbb{Z}$	braiding (c_1)	K3, abelian surface
3	E_1	0	trivial	$K3 \times E$, conifold
4	E_1	$\mathbb{Z}$	shifted symplectic (p_1)	$K3 \times K3$

The complex Bott group $\pi_d(BU) = KU^{-d}$ is 2-periodic:

$$\pi_d(BU) = \begin{cases} \mathbb{Z}, & d \text{ even}, \\ 0, & d \text{ odd}. \end{cases}$$

The 8-periodicity belongs to the real and symplectic refinements $\pi_d(BO)$ and $\pi_d(B\mathrm{Sp})$. In the Vol III effective obstruction tower the trivial dimensions occur at d mod 8 $\in \{1,3,7\}$, with a separate $\mathbb{Z}/2$ symplectic refinement at $d \equiv 5 \pmod 8$.

3 The CY-to-chiral functor

3.1 The two-stage factorisation and its four-step realisation

On the verified loci, the CY-to-chiral assignment factorises as

$$\Phi_d^{(\Sigma_{d-1},C)} = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}} \colon \mathrm{CY}_d^{\mathrm{wit}}\text{-Cat} \longrightarrow E_n\text{-ChirAlg}.$$

Stage 1 produces the canonical E_d-holomorphic factorisation algebra $\Phi_d^{\mathrm{FA}}(\mathcal{C})$ on the CY_d variety X after the relevant formality and holomorphic-locality witnesses have been fixed. Kontsevich–Tamarkin E_d-formality acts on Hochschild cochains with their degree--1 Gerstenhaber bracket; after CY duality to homology the shifted bracket has degree $1-d$. Stage 2 takes the factorisation homology of $\Phi_d^{\mathrm{FA}}(\mathcal{C})$ over a closed $(d-1)$-cycle $\Sigma_{d-1} \subset X$ and restricts to a reference curve $C \subset X$. A single CY_d category thus admits a family of E_1-chiral shadows indexed by (Σ_{d-1}, C).

The four-step construction below is the computational incarnation of the composite $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}$ along a codimension-one cycle followed by curve restriction; each step consumes a specific piece of the CY data and produces a specific algebraic structure on the reference curve C.

Construction 3.1 (The functor Φ). Let $\mathcal{C}$ be a smooth proper CY_d category equipped with the negative-cyclic CY class, formality, holomorphic-twist, framing, completion, and specialisation witnesses required in the dimension under discussion. Let C be the reference curve.

1. **Cyclic A_∞ → Lie conformal algebra.** The Gerstenhaber bracket on $\mathrm{HH}^\bullet(\mathcal{C})$ is a graded Lie bracket of degree -1 (Proposition 2.4). The CY pairing promotes this to a Lie conformal algebra $\mathfrak{L}_\mathcal{C}$: the bracket becomes a λ-bracket $\{a_\lambda b\}$, and the pairing becomes the invariant bilinear form. This step consumes the A_∞-structure and the CY trace; it produces the current algebra of $\mathcal{C}$.
2. **Lie conformal algebra → factorization envelope.** The factorization envelope construction produces a factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_\mathcal{C})$ on any smooth curve X. This step consumes the Lie conformal algebra and the curve; it produces a cosheaf on $\mathrm{Ran}(X)$ whose sections are the OPE data. The universal enveloping vertex algebra $U_X^{\mathrm{fact}}(\mathfrak{L}_\mathcal{C})$ is the chiral algebra.
3. **$\mathbb{S}^d$-framing → E_n enhancement.** The framing witness on $\mathrm{HH}_\bullet(\mathcal{C})$ (Proposition 2.5) enhances the factorization algebra to an E_n-chiral algebra. At $d = 2$: the $\mathbb{S}^2$-framing of Kontsevich–Vlassopoulos gives an E_2-chiral algebra. At $d \geq 3$: the native E_d-structure restricts to E_1 for CY quantum group purposes (the direct E_d-braiding is over-symmetric for braided chiral purposes; the ordered E_1 direction is retained on C).
4. **Quantization.** The negative-cyclic CY class determines the quantization datum for the classical factorization algebra; its Hochschild trace is only the decategorified shadow. The quantum chiral algebra $A_\mathcal{C} = \Phi(\mathcal{C})$ is the output. At $d = 2$, the Serre functor $\mathbb{S}_\mathcal{C} \simeq [2]$ kills the one-loop anomaly by parity, so the quantization preserves κ_{ch}. At $d = 3$, the Serre functor $\mathbb{S}_\mathcal{C} \simeq [3]$ no longer guarantees this; quantum corrections may shift κ_{ch}.

3.2 Theorem CY-A: the $d = 2$ case

Theorem 3.2 (CY-to-chiral functor at $d = 2$). *There exists a functor* $\Phi \colon \mathrm{CY}_2\text{-Cat} \to E_2\text{-ChirAlg}$ *from smooth proper CY categories of dimension 2 to E_2-chiral algebras, such that:*

(i) $\Phi(\mathcal{C})$ *is a chiral algebra whose generating fields are in bijection with* $\mathrm{HH}^{\bullet+1}(\mathcal{C})$.

(ii) The bar complex $B(\Phi(C))$ is quasi-isomorphic to the cyclic bar complex $\mathrm{CC}_\bullet(C)$ as a factorization coalgebra.

(iii) The modular characteristic is the chiral trace supertrace

$$\kappa_{\mathrm{ch}}(\Phi(C)) = \Xi(C).$$

For $C = D^b(\mathrm{Coh}(X))$ with $h^{1,0}(X) = 0$,

$$\Xi(X) = \sum_q (-1)^q h^{0,q}(X) = \chi(\mathcal{O}_X).$$

The full Hochschild Euler characteristic $\sum_i (-1)^i \dim \mathrm{HH}_i(C)$ is a different invariant.

Proof. The first two parts of Construction 3.1 are standard: the Gerstenhaber bracket produces a Lie conformal algebra, and the factorization envelope produces a chiral algebra. The framing part at $d = 2$ uses the Kontsevich–Vlassopoulos $\mathbb{S}^2$-framing on $\mathrm{HH}_\bullet(C)$, which gives an E_2-algebra structure on the cyclic homology; this enhances the factorization algebra to an E_2-chiral algebra. The CY trace supplies quantization; at $d = 2$, the Serre functor $\mathbb{S}_C \simeq [2]$ forces the holomorphic anomaly to vanish on the stated Hodge-filtered trace lane. Part (i) follows from the construction; part (ii) from the identification of the bar differential with the cyclic boundary operator; part (iii) from the Hodge-filtered genus-one coefficient. □

Remark 3.3 (Verification at $d = 2$). For K3 surfaces: $\Xi(D^b(\mathrm{Coh}(S))) = \chi(\mathcal{O}_S) = 2$, and $\kappa_{\mathrm{ch}} = 2$ is verified by six independent paths (direct Hodge, Mukai pairing, Dolbeault contracting homotopy, bar computation, Borcherds lift, elliptic genus). For abelian surfaces: $\kappa_{\mathrm{ch}} = \chi(\mathcal{O}_A) = 0$.

Conjecture 3.4 (Compact Hodge-trace lane at $d \geq 3$). *For $C = D^b(\mathrm{Coh}(X))$ on a constructed compact Hodge-trace lane with $d \geq 3$,*

$$\kappa_{\mathrm{ch}}(\Phi(C)) \;=\; \Xi(X) = \chi(\mathcal{O}_X).$$

This conjecture does not identify the compact Hodge supertrace with the Heisenberg–Mukai specialisation or with the BKM weight.

Remark 3.5 (Evidence). At $d = 3$, compact total-space values such as $\Xi(K3 \times E) = 0$ are distinct from the Heisenberg–Mukai value 3 and the BKM value 5. Non-compact toric local models require their own normalization. The obstruction to extending the $d = 2$ proof is that the $d = 3$ quantization step may introduce corrections: Serre duality $\mathbb{S}_C \simeq [3]$ no longer kills the one-loop anomaly by parity.

3.3 The five objects in the CY context

Under the functor Φ, the five-object chain of Volume I becomes a chain of CY invariants. The five objects are produced by four distinct functors and should never be conflated.

Remark 3.6 (Five objects from $\Phi(C)$). Let $A_C = \Phi(C)$ be the chiral algebra of a CY_d category C.

(i) A_C (the chiral algebra): the factorization envelope of the Lie conformal algebra $\mathfrak{L}_C$.

(ii) $B(A_C) = T^c(s^{-1}\bar{A}_C)$ (the bar coalgebra): a factorization coalgebra on $\mathrm{Ran}(X)$ with deconcatenation coproduct. At genus $g \geq 1$, the fiberwise differential satisfies $d_{\mathrm{fib}}^2 = \kappa_{\mathrm{ch}}(A_C) \cdot \omega_g$.

(iii) For a chosen quadratic presentation $A_C = T_X(V_C)/(R_C)$, the presentation coalgebra is $A^{i}_{C} = C_X(s^{-1}V_C, s^{-2}R_C)$, with $q_{A_C}\colon A^{i}_{C} \to B(A_C)$. On the Koszul locus, $H^\bullet(q_{A_C})$ identifies its cohomology with $H^\bullet B(A_C)$. A BPS comparison datum may then identify the latter coalgebra with the corresponding BPS state space.

(iv) Verdier duality forms $K_X(A_C) = D_{\mathrm{Ran}} B(A_C)$, and a chosen pair map $\nu_{A_C}\colon K_X(A_C) \to A^{!}_{C}$ reaches the Koszul partner. For CY_2 categories, the mirror statement is the corresponding CY-to-chiral comparison theorem.

(v) $Z^{\mathrm{der}}_{\mathrm{ch}}(A_C)$ (the derived chiral center): the chiral Hochschild cochain complex encoding the algebraic closed-sector slot; a physical bulk interpretation requires the open–closed comparison datum.

The bar complex is the E_1 engine; the derived center is the $\mathrm{SC}^{\mathrm{ch,top}}/E_3$ output. The counit $\Omega(B(A_C)) \to A_C$ performs reconstruction, Verdier duality forms $K_X(A_C)$, and categorical Hochschild cochains form the derived center.

4 The $\kappa_\bullet$-spectrum

4.1 Four modular characteristics

A CY manifold X admits multiple chiral algebraizations, each with its own modular characteristic. The four approved subscripts in this volume are: κ_{cat} (categorical Hodge supertrace), κ_{ch} and its explicit specialisations such as $\kappa^{\mathrm{Heis}}_{\mathrm{ch}}$ (boundary chiral trace lanes), κ_{BKM} (Borcherds–Kac–Moody), and κ_{fiber} (lattice). An unsubscripted κ has no invariant meaning on this surface.

Definition 4.1 (The $\kappa_\bullet$-spectrum). For a CY_d manifold X (or more generally a CY_d category $\mathcal{C}$), the *$\kappa_\bullet$-spectrum* is the ordered tuple

$$\mathrm{Spec}_{\kappa_\bullet}(X) = (\kappa_{\mathrm{cat}},\ \kappa^{\mathrm{Heis}}_{\mathrm{ch}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{fiber}})$$

where:

(i) $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_X)$ (the holomorphic Euler characteristic) measures the total-space categorical Hochschild datum. On products it is Künneth-multiplicative; fibre multipliers such as $\chi(\mathcal{O}_{K3}) = 2$ must be named on the fibre lane, not imported into $\kappa_{\mathrm{cat}}(K3 \times E)$.

(ii) $\kappa^{\mathrm{Heis}}_{\mathrm{ch}}$ is the Heisenberg–Mukai boundary specialisation. For $K3 \times E$ it is $3 = \dim_{\mathbb{C}}(K3 \times E)$. The compact total-space chiral trace lane is instead $\kappa_{\mathrm{ch}}(A_{K3\times E}) = \Xi(K3 \times E) = 0$. For $d = 2$ and $h^{1,0} = 0$, Theorem 3.2 gives $\kappa_{\mathrm{ch}} = \chi(\mathcal{O}_X)$.

(iii) $\kappa_{\mathrm{BKM}} = c_N(0)/2$ is the Borcherds weight of the automorphic BKM denominator in the Vol III normalization. For $K3 \times E$, the primitive denominator is Δ_5 and $\kappa_{\mathrm{BKM}} = 5$; the squared unnormalised Igusa convention is $\Phi^{\mathrm{un}}_{10} = \Delta_5^2$.

(iv) $\kappa_{\mathrm{fiber}} = \mathrm{rank}(\Lambda_X)$ (the lattice rank) measures the frame data. For $K3 \times E$: $\kappa_{\mathrm{fiber}} = 24$.

The four characteristics are generically distinct. Their relations encode the internal structure of the CY manifold.

Proposition 4.2 (Spectrum relations for $K3 \times E$). *For $X = K3 \times E$:*

$$\mathrm{Spec}_{\kappa_\bullet}(K3 \times E) = (0,\ 3,\ 5,\ 24).$$

Here the first entry is the total-space value $\chi(\mathcal{O}_{K3\times E}) = 0$, while the value 2 belongs only to the K3 fibre. No additive identity equates the Borcherds weight with the sum of the categorical and chiral entries. The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 5/3$ reflects the single-copy-to-second-quantization passage between the Heisenberg specialisation and the Borcherds denominator.

Proof. $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{K3\times E}) = \chi(\mathcal{O}_{K3})\chi(\mathcal{O}_E) = 2 \cdot 0 = 0$ by Künneth multiplicativity for the total space. $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = \dim_{\mathbb{C}}(K3 \times E) = 3$ (additivity under Cartesian product, Route B: $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3) = 2$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(E) = 1$; verified by six independent paths). This is the Heisenberg-level additive invariant; the total-space Hodge-supertrace Route A $\Xi(X) = \chi(\mathcal{O}_X)$ is Künneth-multiplicative and vanishes on $K3 \times E$. The Volume III Beauville-route comparison separates these two invariants: Route A is the total-space Hodge supertrace, while Route B is the additive Heisenberg-level shadow. $\kappa_{\mathrm{BKM}} = c_1(0)/2 = \mathrm{wt}(\Delta_5) = \frac{1}{2}\,\mathrm{wt}(\Phi_{10}^{\mathrm{un}}) = 5$ because $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ has weight 10. $\kappa_{\mathrm{fiber}} = \mathrm{rank}(H^*(K3,\mathbb{Z})) = 24$ (the second Betti number $b_2(K3) = 22$ plus $b_0 + b_4 = 2$). The numerical decomposition $5 = 2 + 3$ uses the K3 fibre value 2 and the Heisenberg specialisation 3; it is not an identity among total-space $\kappa_\bullet$-invariants. The fibre rank relation $24 = 12 \cdot 2$ uses $\chi(\mathcal{O}_{K3}) = 2$, not $\chi(\mathcal{O}_{K3\times E})$. □

4.2 Koszul complementarity in the CY context

Volume I defines the Koszul conductor $K(A) = \kappa_{\mathrm{ch}}(A) + \kappa_{\mathrm{ch}}(A^!)$: it is 0 for KM/Heisenberg/lattice/free families, 13 for Virasoro, and family-dependent in general. The CY context introduces a new complementarity: mirror symmetry exchanges $\mathcal{C}$ and $\mathcal{C}^\vee$, and $A_{\mathcal{C}^\vee} = A^!_{\mathcal{C}}$ when both are constructed.

Conjecture 4.3 (CY Koszul complementarity). *For a CY$_2$ category $\mathcal{C}$ with mirror $\mathcal{C}^\vee$:*

$$\kappa_{\mathrm{ch}}(\Phi(\mathcal{C})) + \kappa_{\mathrm{ch}}(\Phi(\mathcal{C}^\vee)) = 0.$$

The Koszul conductor vanishes for all mirror pairs at $d = 2$.

Remark 4.4 (Scope). The quintic threefold is a CY$_3$ example and does not supply evidence for the CY$_2$ complementarity statement. Moreover $\chi(\mathcal{O}_{X_5}) = 0$ and $\chi_{\mathrm{top}}(X_5) = -200$, so $\chi(X_5)/2$ is not the categorical trace invariant used here. The CY$_3$ BCOV and mirror lane is a separate problem.

4.3 Shadow depth classification

The CY-to-chiral functor assigns to each CY geometry a chiral algebra whose shadow class (Volume I) organizes the landscape.

Proposition 4.5 (Shadow depth of CY chiral algebras). *Under Φ, the shadow depth classification of Volume I (G/L/C/M) distributes over the CY landscape as follows:*

CY geometry	*Shadow class*	κ_{ch}	*Shadow depth*
$\mathbb{C}^3$ *(toric, abelian)*	*G*	1	2
Resolved conifold	*G*	1	2
Local $\mathbb{P}^2$	*M*	3/2	∞
$K3 \times E$	*M*	3	∞

Classes G and M are populated. Classes L (shadow depth 3) and C (shadow depth 4) are targets of the toric programme: the toric vertex bound $r_{\max} \leq N$ (where N is the number of web-diagram vertices) predicts *that specific toric geometries should realize these classes, but explicit CY examples at shadow depth exactly* 3 *or* 4 *remain to be identified.*

5 The $K3 \times E$ prototype

The product of a K3 surface S and an elliptic curve E is the canonical CY_3 testing ground. It is simple enough to compute (lattice $\Lambda^{3,2}$ of signature $(3, 2)$, K3 elliptic genus $\phi_{0,1}(\tau, z)$, Borcherds denominator Δ_5) and already exhibits every obstruction that separates the shadow tower from the full automorphic form.

5.1 The thirteen structural results

The structural spine begins with six computations: modular characteristic, moonshine multiplier, BKM weight, BPS factorisation, mock modular decomposition, and shadow convergence.

K3-1 (Heisenberg-level modular characteristic, Route B): $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3 \times E) = 3 = \dim_{\mathbb{C}}$, verified by six independent paths (direct Hodge, Mukai pairing, Dolbeault contracting homotopy, bar computation, Borcherds lift, elliptic genus). Additivity under Cartesian product: $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3) = 2$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(E) = 1$. The Route A Hodge-supertrace $\Xi(X) = \chi(\mathcal{O}_X)$ is Künneth-multiplicative and gives $\Xi(K3 \times E) = 2 \cdot 0 = 0$; the two routes label genuinely distinct invariants. The Volume III Beauville-route comparison records the same split between total-space trace and Heisenberg-level additive shadow.

K3-2 (moonshine multiplier): the factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is $\kappa_{\mathrm{cat}}(K3) = \chi(\mathcal{O}_{K3}) = 2$. The massive Mathieu multiplicities satisfy $A_n = \kappa_{\mathrm{cat}} \cdot \dim(\rho_n)$: the bar complex provides the universal coefficient and does not detect M_{24} directly.

K3-3 (BKM weight): $\kappa_{\mathrm{BKM}} = \frac{1}{2}\,\mathrm{wt}(\Phi_{10}^{\mathrm{un}}) = 5$. The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ reflects second quantization; the Hilbert–Chow exceptional divisor accounts for $\chi(\mathrm{Hilb}^2) - \chi(\mathrm{Sym}^2) = 24 = \chi(K3)$.

K3-4 (BPS factorization): $A_n = \kappa_{\mathrm{cat}}(K3) \cdot \dim(\rho_n)$, separating the fibre homological part ($\kappa_{\mathrm{cat}}(K3) = 2$) from the group-theoretic part (ρ_n, the Mathieu representation).

K3-5 (mock modular verification): the mock modular decomposition $Z_{K3} = (h - 24\mu)\,\vartheta_1^2/\eta^3$ is verified to machine precision ($\sim 5.7 \times 10^{-16}$ relative error), where μ is the Zwegers Appell–Lerch sum.

K3-6 (shadow convergence): the shadow generating function $Z^{\mathrm{sh}}(t) = \kappa_{\mathrm{ch}}\big((t/2)/\sin(t/2) - 1\big)$ converges with radius $R = 2\pi$, has entire Borel transform, and exhibits no resurgence.

5.2 The shadow–Siegel gap theorem

Theorem 5.1 (Shadow–Siegel gap). *The shadow obstruction tower of $K3 \times E$ does not produce the unnormalised Igusa cusp form Φ_{10}^{un}. Four structural obstructions separate the two:*

(O1) Categorical*: the shadow tower produces numbers (Taylor coefficients of Θ_{A_C}); the denominator identity is a function (an automorphic form on $\mathcal{H}_2$).*

(O2) $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}$–$\kappa_{\mathrm{BKM}}$ mismatch*:*

$$\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3 \neq 5 = \kappa_{\mathrm{BKM}}.$$

The compact total-space Hodge-supertrace value is separately $\kappa_{\mathrm{ch}}(A_{K3\times E}) = 0$. The number 3 *is the Heisenberg–Mukai specialisation, not the total-space Φ-trace.*

(O3) Second quantization*: the physical DT partition function is the DMVV symmetric product of single-copy data; the bar complex of a single algebra is the single copy.*

(O4) Schottky: *at $g \geq 4$, the Torelli map $\mathcal{M}_g \to \mathcal{A}_g$ is no longer dominant, and the shadow tower is imprisoned in tautological classes on $\overline{\mathcal{M}}_g$.*

Each obstruction points to a research programme. The $\kappa_{\mathrm{ch}} - \kappa_{\mathrm{BKM}}$ mismatch (O2) is the most telling: a single CY_3 admits multiple chiral algebraizations, and the $\kappa_\bullet$-spectrum records which face of the geometry each algebraization captures.

6 Toric CY_3 and the cohomological Hall algebra

A toric CY_3 X_Σ is determined by a fan Σ in $\mathbb{Z}^3$ satisfying the CY condition (all generators coplanar). The toric diagram fixes a quiver with potential (Q_X, W_X), and the critical CoHA $\mathcal{H}(Q_X, W_X) = \bigoplus_{\mathbf{d}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$ provides the E_1-sector of the CY quantum group.

6.1 $\mathbb{C}^3$ and the affine Yangian

The simplest toric CY_3. Toric diagram: a single triangle. Quiver: the Jordan quiver (one vertex, one loop $\circlearrowleft$). Cubic potential $W = \mathrm{tr}(XYZ - XZY)$.

Theorem 6.1 (Schiffmann–Vasserot). *The critical CoHA of $\mathbb{C}^3$ is isomorphic to the positive half of the affine Yangian of $\mathfrak{gl}_1$:*

$$\mathrm{CoHA}(\mathbb{C}^3) \;\simeq\; Y^+(\widehat{\mathfrak{gl}}_1).$$

The full $\mathcal{W}_{1+\infty}$ vertex realisation appears only after taking the appropriate double/evaluation realisation.

The chain $\mathbb{C}^3 \to Y^+(\widehat{\mathfrak{gl}}_1) \to \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ is verified end-to-end: the toric CY_3 produces the CoHA, which is the positive half of the Yangian and hence the E_1-chiral quantum group. The $\mathcal{W}_{1+\infty}$ vertex realisation enters only after the double/evaluation step. The Drinfeld center recovers the full E_2-braiding (Theorem 6.3 below).

6.2 The CoHA as ordered Hall input

The CoHA is an associative (E_1) algebra: the Hall multiplication is ordered (short exact sequences $0 \to V' \to V \to V'' \to 0$ have sub before quotient). In the present framework:

- The CoHA is the positive half: an ordered associative Hall algebra before chiralisation.
- The full quantum vertex realisation is obtained only after the double/center construction, where the E_2-braiding lives.
- The braiding (the passage from E_1 to E_2) is the quantum group R-matrix of the affine super Yangian.
- The ordered bar $B^{\mathrm{ord}}(A_C)$ carries the collision kernel and its represented R-operator. The symmetric bar $B^\Sigma(A_C)$ carries the homotopy-coinvariant operator $q_2(K^{\mathrm{coll}}_{A_C})$. A normalized functional $\tau^{\mathrm{res}}_{A_C,2}$ gives κ_{ch} in abelian and scalar families and κ_{dp} on the non-abelian affine row; the Sugawara contribution $\dim(\mathfrak{g})/2$ completes κ_{ch} there.

For local toric charts the hCS BV complex and the Hall shuffle algebra glue at finite positive-half Rees level: after truncating the Rees filtration, the positive-half Hall multiplication and the hCS extension correspondence define the same ordered E_1 operation. Vol III now constructs this passage on the resolved conifold by gluing the hCS E_3 factorisation algebra, adding the anomaly-free quantum lift, and writing the finite hCS–Rees-Hall chart maps. The compact Hall, quasi-NCCR, and recognised Hall-double comparisons are separate global identifications; finite radical-quotient doubles now exist from compact windows, while the Borcherds recognition/completion remains a finite-defect and inverse-limit condition.

Remark 6.2 (Shadow = GW($\mathbb{C}^3$)). At $\kappa_{\mathrm{ch}} = \Psi$ (the deformation parameter of $\mathcal{W}_{1+\infty}$ at $c = 1$), the shadow obstruction tower recovers the genus-1 constant-map channel for $\mathbb{C}^3$. The all-genus virtual identification requires the uniform-weight and $\Phi_3^{(\Sigma_2, C)}$ hypotheses, and the motivic identification remains conjectural. On the DT side, the MacMahon function $M(q) = \prod_{n \geq 1}(1 - q^n)^{-n}$ encodes the partition function as the second-quantized bar/DT character in the non-compact degree-zero normalization.

6.3 The Drinfeld center

Theorem 6.3 (Drinfeld center for $\mathbb{C}^3$). *The Drinfeld center of the E_1-monoidal representation category of $Y^+(\widehat{\mathfrak{gl}}_1)$ recovers the full E_2-braided monoidal structure:*

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \simeq \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)) \simeq \mathrm{Rep}^{E_2}(\mathcal{W}_{1+\infty}).$$

Five invariants match at six levels: the ordinary center has dimension 1*, while the derived center has dimension* 3 *(capturing* $\mathrm{Ext}^{1,2}$ *invisible to the commutant).*

The Drinfeld center is the categorical-centre step in the Volume I circle: it is the right adjoint to forgetting the braiding and recovers non-trivial half-braidings at the categorical level. The averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$ is a different, lossy projection from ordered R-matrix data to the scalar κ_{ch} shadow in the $\kappa_\bullet$-spectrum.

7 The E_1 stabilization theorem

Theorem 7.1 (E_1 stabilization for toric local models). *For toric $GL(d)$-invariant local CY models with $d \geq 3$, the Stage-2 CY chiral algebra $A_C = \Phi(C)$ is natively E_1 on the reference curve. In this local invariant lane the selected CY polyvector sector has trivial Schouten–Nijenhuis bracket, and the CoHA extension correspondence imposes a preferred ordering (sub before quotient). For general CY categories the full Schouten–Nijenhuis bracket remains nonzero; the E_1 level is forced by Stage 2 and degree. The braided monoidal structure is recovered through the Drinfeld center.*

Proof. The Schouten–Nijenhuis bracket on $\mathrm{PV}(\mathbb{C}^d)$ is nonzero in general; vector fields already bracket. The toric local model uses the $GL(d)$-invariant CY sector generated by the volume class and the scalar torus-weight data entering the Hall construction. On this selected sector the Schouten output has no invariant target of the required CY weight, hence the induced bracket is zero. The quantum operation then comes from the CoHA convolution correspondence, which is associative (E_1) and ordered by subobject before quotient. □

Remark 7.2 (Arithmetic of the fundamental group). The key arithmetic for ordered configurations is

$$\pi_1(\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{R}^2)) = \mathbb{Z}, \qquad \pi_1(\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{R}^d)) = 0 \quad (d \geq 3).$$

For the unordered quotient,

$$\pi_1(\mathrm{Conf}_2^{\mathrm{unord}}(\mathbb{R}^d)) = \mathbb{Z}/2 \quad (d \geq 3).$$

Thus the native E_d operation at $d \geq 3$ is over-symmetric for braided chiral purposes; the construction keeps the ordered E_1 direction on C and recovers braiding through the Drinfeld center.

8 Φ scope, CY-dimension stratification, Borcherds weight

Three consolidated structural theorems pin the CY-to-chiral assignment to its precise scope: the native E_n-output level on the reference curve, the dimension-stratified trace spectrum, and the Vol III Borcherds weight normalisation $\kappa_{\mathrm{BKM}} = c_N(0)/2$.

Theorem 8.1 (Native E_n-output scope of the two-stage $\Phi_d^{(\Sigma_{d-1},C)}$). *Let*

$$\Phi_d^{(\Sigma_{d-1},C)} = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}$$

be evaluated on a constructed locus with the required formality, holomorphic-twist, framing, completion, and specialisation witnesses. Its native output level is

$$n = \begin{cases} \infty & d = 1, \\ 2 & d = 2, \\ 1 & \text{constructed } d \geq 3. \end{cases}$$

At $d = 2$ the native $\mathbb{S}^2$-framing on $\mathrm{HH}_\bullet(C)$ supplies a genuine E_2-chiral structure (Kontsevich–Vlassopoulos; Theorem CY-A$_2$). At $d = 3$ this is a witnessed object-level assignment for fixed (Σ_2, C); arbitrary functoriality on all CY morphisms is not asserted. For $d \geq 4$ the display names a construction problem, not a theorem. Braiding is recovered categorically via the Drinfeld centre, closing the E_n operadic circle on $Z^{\mathrm{der}}_{\mathrm{ch}}$ rather than on a bare output.

Theorem 8.2 (Trace stratification by CY-dimension). *For a compact Calabi–Yau manifold X of dimension d and a reference curve $C \subset X$ with specialisation cycle $\Sigma_{d-1} \subset X$,*

$$\kappa_{\mathrm{ch}}^{\mathrm{Hodge}}\big(\Phi_d^{(\Sigma_{d-1},C)}(D^b\mathrm{Coh}(X))\big) = \sum_{q=0}^{d} (-1)^q h^{0,q}(X),$$

on the compact Hodge-trace lane. The stratification across the canonical families reads

family (dim d)	$\kappa_{\mathrm{ch}}^{\mathrm{Hodge}}$	*notes*
elliptic curve E ($d = 1$)	0	$h^{0,0} - h^{0,1} = 1 - 1$
$K3$ ($d = 2$)	2	$\chi(\mathcal{O}_{K3}) = 2$
abelian/bielliptic surface ($d = 2$)	0	$\chi(\mathcal{O}) = 0$
quintic $\subset \mathbb{P}^4$ ($d = 3$)	0	$\chi(\mathcal{O}_X) = 0$
$K3 \times E$ ($d = 3$)	0	$\chi(\mathcal{O}) = 0$
triple elliptic E^3 ($d = 3$)	0	$\chi(\mathcal{O}) = 0$
local $\mathbb{P}^2 = \mathrm{Tot}(K_{\mathbb{P}^2})$	$3/2$	$\chi(S)/2 = 3/2$ *(shadow)*
CY_4 *sextic $\subset \mathbb{P}^5$*	2	*Hodge supertrace*
generic CY_5	0	*Hodge supertrace vanishes*

The Heisenberg–Mukai, BKM, and fibre lanes are not entries of this table. The conifold is not *a local surface: its geometry is* $\mathrm{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1)$*, not* $\mathrm{Tot}(K_S \to S)$*, and the identity* $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = \chi(S)/2$ *does not apply. The McKay orbifold* $[\mathbb{C}^3/\mathbb{Z}_3]$ *gives* $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = 1$ *by direct calculation.*

Proof. The identification $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = \sum(-1)^q h^{0,q}$ is the Hochschild–Kostant–Rosenberg degeneration composed with the Mukai pairing; the degree-d cyclic trace $\mathrm{Tr}\colon \mathrm{HC}_d^-(C) \to k$ restricts to the top-dimensional Hodge piece, and Serre duality pairs it with the full Hodge decomposition. The family entries are direct computation; "local $\mathbb{P}^2$" comes from the shadow $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = \chi(S)/2 = 3/2$ because $\chi(\mathbb{P}^2) = 3$. The conifold exclusion is structural: its total space is not the total space of a canonical line bundle over a surface, so the local-surface formula is inapplicable. □

Theorem 8.3 (Borcherds weight normalisation)**.** *On the Vol III Borcherds-denominator locus* $N \in \{1, 2, 3, 4, 6\}$,

$$\kappa_{\mathrm{BKM}}(\Phi_N) = \frac{c_N(0)}{2}.$$

Thus $c_N(0) = (10, 6, 4, 3, 2)$ (*Jatkar–Sen / Govindarajan–Krishna; the fourth rung half-integral*) *gives*

$$\kappa_{\mathrm{BKM}} = (5, 3, 2, \tfrac{3}{2}, 1).$$

This is a Borcherds-weight identity. The compact Hodge supertrace is separate, and the ghost-charge normalisation belongs to the Monster–Leech lane.

Remark 8.4 (κ subscript conventions (all subscripts explicit)). Every κ in this section carries an approved Vol III subscript from the closed set {ch, cat, BKM, fiber}. The identifications $\kappa_{\mathrm{ch}} \to \chi(\mathcal{O})$ (Hodge supertrace), $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} \to \dim_{\mathbb{C}} X$ on the Heisenberg–Mukai specialisation, and $\kappa_{\mathrm{BKM}} \to c_N(0)/2$ are the precise uses here.

Remark 8.5 (Dimension range of $\Phi_d^{(\Sigma_{d-1}, C)}$). Stage 1 Φ_d^{FA} is unconditional for $d \in \{1, 2\}$ and is constructed at $d = 3$ on the witnessed filtered locus by Kontsevich–Tamarkin E_d-formality applied to Hochschild cochains, Costello–Gwilliam–Li locality on X, and the required framing and completion witnesses. Stage 2 $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1}, C}$ is factorisation homology followed by curve restriction along a witnessed admissible specialisation datum (Σ_{d-1}, C). $\Phi_d^{(\Sigma_0, C)}$ at $d = 1$: trivial (A_∞ = comm). $d = 2$: proved (Theorem CY-A$_2$), E_2-chiral output. $d = 3$: witnessed object-level output for fixed (Σ_2, C), E_1-chiral on the reference curve; the family of constructed specialisations $\{\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_2, C}(\Phi_3^{\mathrm{FA}}(C))\}_{\Sigma_2}$ over choices of $K3$-, T^4-, or Enriques-fibres produces the Borcherds–Δ_5, Igusa–χ_{10}, and Gritsenko–Cléry paramodular shadows respectively. $d = 4$: Stage 1 is obstructed by $\pi_4(BU) = \mathbb{Z}$; the p_1-twisted double current algebra is the surviving structure. For $K3 \times K3$ one has $\int_{K3 \times K3} p_1(T_X)^2 = 4608$, so this product lies on the p_1-twisted stratum rather than on a Pontryagin-vanishing locus. The untwisted E_4 lift requires separate vanishing data. For $d = 5, 6$ no general Stage 1 theorem is asserted here; the expected Hodge-supertrace value $\kappa_{\mathrm{ch}} = 0$ for a generic smooth CY requires a separate existence theorem for the relevant higher factorisation algebra. Thus the unconditional range is $d \in \{1, 2\}$, the constructed $d = 3$ range is witnessed object-level, and the $d = 4$ untwisted range is the Pontryagin-vanishing family; p_1-twisted examples such as $K3 \times K3$ require the twisted datum.

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